Lab: Motor Circuits and Action

> **Learning Goal:** Trace motor control circuits and analyze disorders through the Active Inference lens.

Part 1: The Basal Ganglia Loop

Exercise: Complete the diagram of the cortico-basal ganglia-thalamo-cortical loop:

1. Draw or describe the five steps: Cortex → Striatum → Direct/Indirect pathway → Globus pallidus → Thalamus → back to Cortex
2. Label where dopamine acts and its effect on each pathway
3. Predict what happens when: (a) dopamine is high (b) dopamine is low (c) dopamine is absent

Write your response here

Part 2: Motor Predictions vs. Motor Commands

> **Learning Goal:** Contrast the traditional and Active Inference views of motor control.

Scenario: You reach for a glass of water and bring it to your lips.

Step	Traditional "Command" View	Active Inference "Prediction" View
Planning		
Initiation		
Execution		
Error correction		
Completion		

Fill in both columns to contrast the two perspectives.

Write your response here

Part 3: Cerebellar Learning Simulation

> **Learning Goal:** Trace the error-correction circuit in the cerebellum.

Exercise: You're learning to throw darts at a target.

1. First throw: You aim for center, the dart lands far right. What is the climbing fiber error signal?
2. How does this error modify the parallel fiber–Purkinje cell synapses?
3. After 100 throws: What has happened to the prediction error? What has happened to cerebellar predictions?
4. If you suddenly switch to a heavier dart, what happens? How does the cerebellum adapt?

Write your response here

Part 4: Motor Disorder Analysis

Learning Goal: Map motor disorders to specific circuit disruptions.

For each disorder, identify the neural structure affected, the Active Inference component disrupted, and one observable symptom:

Disorder	Structure	AI Component	Symptom
Parkinson's disease			
Huntington's disease			
Cerebellar ataxia			
Tourette syndrome			

Write your response here

Part 5: Reflection

In 150 words, reflect: If the motor cortex sends predictions rather than commands, does this change how we think about voluntary vs. involuntary movement? Are these categories real, or are they different points on a precision continuum?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Circuit diagramming	Basal ganglia Go/No-Go pathways
2	Framework comparison	Commands vs. predictions
3	Error-tracing	Cerebellar learning circuit
4	Clinical mapping	Motor disorders and circuit damage
5	Philosophical reflection	Voluntary vs. involuntary action